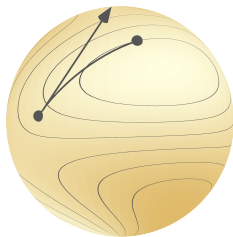




NTNU

Norges teknisk-naturvitenskapelige universitet



Manopt.jl

Numerical Optimization on Manifolds in Julia

Ronny Bergmann

NTNU, Trondheim, Norway.

Geometric Methods in Optimization and Numerical Analysis

Oberwolfach Workshop 2630,

July 21, 2026

Motivation

For a function $f: \mathcal{M} \rightarrow \mathbb{R}$ defined on a Riemannian manifold $\mathcal{M}$ compute

$$p^* \in \arg \min_{p \in \mathcal{M}} f(p)$$

- ▶ easily accessible in math & code
- ▶ interchangeable & configurable algorithms & manifolds
- ▶ efficient & fast

Manopt family.



manoptjl.org

[RB, 2022]



manopt.org

[Boumal, Mishra, Absil, Sepulchre, 2014]



pymanopt.org

[Townsend, Koep, Weichwald, 2016]

A Riemannian Manifold $\mathcal{M}$

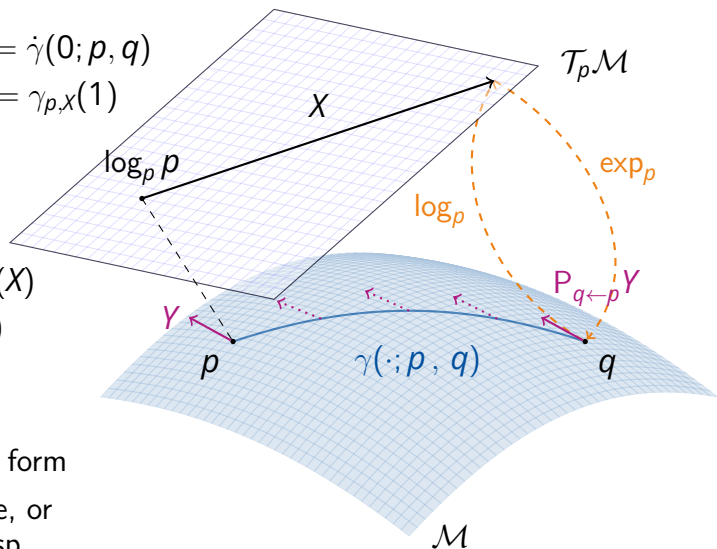
Notation.

- ▶ Logarithmic map $\log_p q = \dot{\gamma}(0; p, q)$
- ▶ Exponential map $\exp_p X = \gamma_{p,X}(1)$
- ▶ Geodesic $\gamma(\cdot; p, q)$
- ▶ Tangent space $\mathcal{T}_p \mathcal{M}$
- ▶ inner product $(\cdot, \cdot)_p$
- ▶ parallel transport $\text{PT}_{p \leftarrow q}(X)$
- ▶ distance function $d(p, q)$

Numerics.

$\exp_p$, $\log_p$, $\text{PT}_{p \leftarrow q}$ maybe not avail, efficiently/in closed form

$\Rightarrow$ use a retraction, its inverse, or a vector transport instead, resp.



Why Julia?

- ▶ high-level language, open source
- ▶ multiple dispatch & operator overloading, for example
`*(::Number, ::AbstractTangentVector)`
allows to write $\alpha * X$ for a tangent vector X
- ▶ just-in-time compilation, solves **two-language problem**
 $\Rightarrow$ “nice to write” and as fast as C/C++

I like the community

- ▶ very welcoming Discourse, Slack, Zulip, ...
- ▶ good package ecosystem organisation
- ▶ packages very modular and relatively easy get them to interact
- ▶ packages: environments, versioning and version resolution

ManifoldsBase.jl – Motivation



Goal. Provide a generic interface to manifolds for

- ▶ defining own (new) manifolds
- ▶ implementing **generic** algorithms on an arbitrary manifold $\mathcal{M}$

Manifold. A Riemannian manifold as subtype of `<:AbstractManifold{F}`

- ▶ $F \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$: field the manifold is build on
- ▶ stores all “general” information, e. g. the manifold dimension
- ▶ example (from `Manifolds.jl`): `M = Sphere(2)`
- ▶ meta manifolds: product $\mathcal{M}_1 \times \mathcal{M}_2$, power $\mathcal{M}^n$, `TangentBundle(M)`

Points and Tangent vectors.

- ▶ by default not typed, often `<:AbstractArray`
- ▶ or use `<:AbstractManifoldPoint` and `<:AbstractTangentVector`

ManifoldsBase.jl – Functions



Goal. Efficient and reusable functions.

Functions. We provide functions like

- ▶ `exp(M, p, X)`, `log(M, p, q)`, `inner(M, p, X, Y)`,
`parallel_transport_to(M, p, X, q)`
- ▶ yields defaults for `norm(M, p, X)` or `shortest_geodesic(M, p, q)`
- ▶ `retract(M, p, X, method)` to approx. $\exp_p X$, with different `methods`,
- ▶ similarly `inverse_retract(M, p, q, m)` and
`vector_transport_to(M, p, X, q, m)`

Efficiency. `exp(M, p, X)` allocates `q` a point and calls `exp!(M, q, p, X)`.

For a new manifold only implement `exp!(M, q, p, X)` to work in-place of `q`
 $\Rightarrow$ avoid memory allocations where possible.

Manopt.jl

Problem and Solver State in Manopt.jl

A problem

`<:AbstractManoptProblem`

containing at least

- ▶ an `AbstractManifold`
- ▶ an `AbstractManifoldObjective`

Example.

```
o = ManifoldGradientObjective(f, g)
problem = DefaultManoptProblem(M, o)
```

The objective can be **decorated**

- ▶ to **cache** function evaluations
- ▶ to **count** function evaluations

A solver state

`<:AbstractManoptSolverState`

to specify/store

- ▶ algorithm parameters, e. g.
 - ▶ a `StoppingCriterion`
 - ▶ a `Stepsize`
- ▶ values like the iterate $p^{(k)}$.

Example.

```
state = GradientDescentState(M)
```

The state can be **decorated**

- ▶ to print **debug**
- ▶ to **record** values

Two methods for a Random Walk solver

```
struct RandomWalkState{P, R, S}
    <: AbstractManoptSolverState
    p::P # current iterate
    q::P # best point visited
    σ::Float64
    rm::R # A retraction method
    stop::S # A StoppingCriterion
end
```

Initialization

```
function initialize_solver!(
    mp::AbstractManoptProblem,
    r::RandomWalkState
) # Set best to init point
    M = get_manifold(mp)
    copyto!(M, r.q, r.p)
    return r
end
```

The k th step

```
function step_solver!(
    mp::AbstractManoptProblem,
    r::RandomWalkState, k
) # generate and scale direction
    M = get_manifold(mp)
    X = rand(M; vector_at=r.p)
    X .*= r.σ/norm(M, r.p, X)
    # Walk
    retract!(M, r.p, r.p, X, r.rm)
    # Check/Store best
    if get_cost(mp, r.p) <
        get_cost(mp, r.q)
        copyto!(M, r.q, r.p)
    end
    return r
end
```

A solver run

Setting up some small example manifold and cost

```
M = Sphere(2) # From Manifolds.jl
data = [[1.0, 0.0, 0.0],[0.0, 1.0, 0.0],[0.0, 0.0, 1.0]]
f(M,p) = sum( distance(M, p, q)^2 for q in data)
```

and a state and problem

```
state = RandomWalkState(
    copy(M, data[1]), # Initial p
    copy(M, data[1]), # Initial q
    0.1, # a value for  $\sigma$ 
    ExponentialRetraction(), # use exp
    StopAfterIteration(1000), # when to stop
)
problem = DefaultManoptProblem(M, ManifoldCostObjective(f))
```

Running the solver: `solve!(problem, state);`

But this is not that easy to use nor comfortable

Ease of use: High-level interfaces

All solvers have a simple function to call for example

```
gradient_descent(M, f, grad_f)           # starts at random point  
gradient_descent!(M, f, grad_f, p)      # starts at p, overwrites p
```

Parameters are passed as keyword arguments. All accept for example

- ▶ `stopping_criterion =`
`StopWhenGradientNormLess(1e-9) | StopAfterIteration(1000)`
Specify when to stop, can be combined with `|` and `&`
- ▶ `debug = [:Iteration, :Cost, "\n", 10, :Stop]`
Print the k and current cost every 10th iteration and reason to stop
- ▶ `record = [:Iterate, :Cost]`
Record the current iterate $p^{(k)}$ and its cost value $f(p^{(k)})$
- ▶ `return_state = false` (default)
return the whole solver state (instead of just the last iterate)
then use `get_record(state)` to access vector of recorded values

The JuliaManifolds ecosystem

We collect several packages in the GitHub organisation [JuliaManifolds](#).

ManifoldsBase.jl An interface for Riemannian manifolds

Manopt.jl Numerical optimization on manifolds

ManoptExamples.jl Reproducible examples of solvers from [Manopt.jl](#)

Manifolds.jl A library of Riemannian manifolds

LieGroups.jl An interface for and a library of Lie groups

ManifoldDiff.jl differentiations of certain function and AD support for functions on manifolds

ManifoldDiffEq.jl use manifolds together [DiffEq.jl](#) to implement solvers for differential equations on manifolds / Lie groups

ManifoldMakie.jl plot manifold-valued data with [Makie.jl](#)

ManifoldAsymptote.jl render manifold-valued data with [Asymptote](#) and a work-in-progress (help wanted!) [ManifoldDistributions.jl](#)

Examples

Riemannian Center of Mass

For $q_i \in \mathcal{M}$, $i = 1, \dots, m$,
consider the
Riemannian Center of Mass

$$f(p) := \frac{1}{2m} \sum_{i=1}^m d^2(p, q_i)$$

with Riemannian gradient

$$\text{grad} f(p) = -\frac{1}{m} \sum_{i=1}^m \log_p q_i$$

```
using Manifolds, Manopt
m = 20; σ = π/8; M = Sphere(2)
p = [0.0, 0.0, 1.0]
Q = [ exp(M, p, σ * rand(M; vector_at=p))
      for _ in 1:m
    ]
f(M, p) = 1/(2m) * sum(
    distance(M, p, q)^2 for q in Q
)
grad_f(M, p) = - 1/m * sum(
    log(M, p, q) for q in Q
)

p1 = gradient_descent(M, f, grad_f, p0)
p2 = gradient_sampling(M, f, grad_f, p0)
      (load RipQP, QuadraticModels for the subsolver)
```

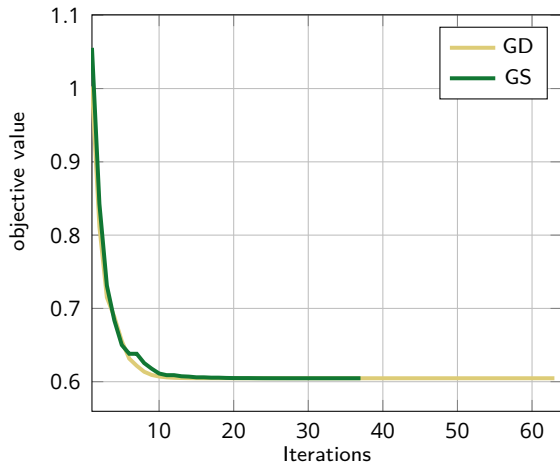
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```
p1 = gradient_descent(M, f, grad_f, p0)
p2 = gradient_sampling(M, f, grad_f, p0)
(load RipQP, QuadraticModels for the subsolver)
```

Riemannian Median

For the Riemannian Median

$$f(p) := \sum_{i=1}^m d(p, q_i)$$

the problem is nonsmooth.

Use the proximal map

$$\text{prox}_{\lambda f}(p) := \arg \min_{q \in \mathcal{M}} d(q, p)^2 + \lambda f(q)$$

for each $f_i(p) = d(p, q_i)$.

Run the

cyclic proximal point algorithm

$$\begin{aligned} p_{k+\frac{i+1}{m}} &= \text{prox}_{\lambda f_i}(p_{k+\frac{i}{m}}), \\ i &= 0, \dots, m-1, \\ k &= 0, 1, \dots \end{aligned}$$

```
using ManifoldDiff, Manifolds, Manopt
using ManifoldDiff: prox_distance
# [...same data Q and p0 as before...]
```

```
f(M, p) = 1/m * sum(
    distance(M, p, q) for q in Q
)
p_f = Function[ #An array of functions
    (N, λ, p) -> prox_distance(N, λ/m, q, p, 1)
    for q in Q
]
```

```
p1 = cyclic_proximal_point(M, f, p_f, p0)
```

Or a derivative-free algorithm: LTMADS

```
p2 = mesh_adaptive_direct_search(M, f, p0)
```

Robust Levenberg-Marquardt

Nonlinear Least Squares

[Baran, RB, 2026]

Goal. Minimize a set $F_i: \mathcal{M} \rightarrow \mathbb{R}^{n_i}$ of **residuals**

$$f(p) = \frac{1}{2} \sum_{i=1}^m \rho_i (\|F_i(p)\|_2^2), \quad p \in \mathcal{M},$$

where $\rho_i: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$, $\rho_i(0) = 0$, are **robustifiers**.

▶ **nonlinear least squares** $\rho(t) = t$

[Adachi, Okuno, Takeda, 2022]

▶ **robust regression** $\rho(t) = \sqrt{t}$

▶ here: **relaxed**, $\rho \in C^1$, e.g. **Huber** $\rho_{\text{Hu}}(t) := \begin{cases} t & \text{if } t \leq 1 \\ 2\sqrt{t} - 1 & \text{if } t > 1. \end{cases}$

Component wise case.

For $F = (f_1, \dots, f_m): \mathcal{M} \rightarrow \mathbb{R}^m$, $f_i: \mathcal{M} \rightarrow \mathbb{R}$ (all $n_i = 1$):
each component is robustified $\Rightarrow$ aims to minimize $\|F(p)\|_1$

The Jacobian and its adjoint

The **Jacobian** of $F: \mathcal{M} \rightarrow \mathbb{R}^n$, $F = (f_1, \dots, f_n)$ is

$$\mathcal{J}_F(p)[X] = DF(p)[X] = \left(\langle \text{grad } f_j(p), X \rangle_p \right)_{j=1}^n, \quad p \in \mathcal{M}, X \in T_p \mathcal{M}.$$

With a basis $Y_1, \dots, Y_d \in T_p \mathcal{M}$: $\mathcal{J}_F(p) = J_F(p) \in \mathbb{R}^{n \times d}$ as a matrix.

But this requires to decompose $X = \sum_i c_i Y_i$ to perform $J_F(p)c$.

The (Riemannian) **adjoint Jacobian** $\mathcal{J}_F^*: \mathbb{R}^n \rightarrow T\mathcal{M}$ is given by

$$\mathcal{J}_F^*(p)[x] = \sum_{j=1}^n x_j \text{grad } f_j(p), \quad p \in \mathcal{M}, x \in \mathbb{R}^n.$$

Again: With an orthonormal basis of $T_p \mathcal{M}$: a matrix $J_F^*(p) = J_F(p)^\top$.

Second order Taylor expansion

For a “block” $G(p) := \frac{1}{2}\rho(\|F(p)\|_2^2)$ the second order Taylor expansion

$$G(R_p(X)) = G(p) + \langle \text{grad } G(p), X \rangle_p + \frac{1}{2} \langle \text{Hess } G(p)[X], X \rangle_p + o(\|X\|_p^3)$$

reads

$$\begin{aligned} G(p) &+ \rho'(\|F(p)\|_2^2) \langle \mathcal{J}_F^*(p)[F(p)], X \rangle_p \\ &+ \rho'(\|F(p)\|_2^2) \left(\langle \mathcal{J}_F^*(p)[\mathcal{J}_F(p)[X]], X \rangle_p + \frac{1}{2} \sum_{j=1}^n f_j(p) \langle \text{Hess } f_j(p)[X], X \rangle_p \right) \\ &+ \rho''(\|F(p)\|_2^2) \langle F(p), \mathcal{J}_F(p)[X] \rangle \langle \mathcal{J}_F^*(p)[F(p)], X \rangle_p + o(\|X\|_p^3) \end{aligned}$$

► drop the cubic error term

► drop the **Hessian terms of F**

⇒ We only require $\rho'(\|F(p)\|_2^2)$, $\rho''(\|F(p)\|_2^2)$, $\mathcal{J}_F$, and $\mathcal{J}_F^*$

Main step of Levenberg-Marquardt

At every iteration $p = p^{(k)}$ find minimizer X^* of the **damped** surrogate

$$s_\lambda(X) := \frac{1}{2} \|\mathcal{L}(X) + y\|_2^2 + \lambda \|X\|_p^2, \quad X \in T_p \mathcal{M}, \lambda \geq 0$$

- ▶ with linear map $\mathcal{L}: T_p \mathcal{M} \rightarrow \mathbb{R}^n$ and $y \in \mathbb{R}^n$.
- ▶ chosen s. t. s_0 approximates G , i.e. with $r = F(p)$, $t = \|r\|^2$ that
 1. $\text{grad } s_0(0_p) = \text{grad } G(p) = \rho'(t) \mathcal{J}_F^*(p)[r]$
 2. $\text{Hess } s_0(0_p)[X] = \rho'(t) \mathcal{J}_F^*(p)[\mathcal{J}_F(p)[X]] + 2\rho''(t) \langle r, \mathcal{J}_F(p)[X] \rangle \mathcal{J}_F^*(p)[r]$.
- ▶ update λ based on “model fit” $m = \frac{G(p) - G(R_p(X^*))}{\frac{1}{2}(s_\lambda(0_p) - s_\lambda(X^*))}$

Overall we have to solve a **sub problem** in every iteration.

Manopt.jl Interlude: Sub solvers

To solve the surrogate sub problem, the `LevenbergMarquardState` stores

- ▶ `sub_problem` for the objective and here the tangent space as manifold
 - ▶ `sub_state` specify
 - ▶ the solver from `Manopt.jl` to use
 - ▶ or that function implements the sub solver (in closed form)
 - ▶ default here `ConjugateResidualState` [Lai, Yoshise, 2024]
 - ▶ `set_parameter!(element, :Name, value)` to update e.g. the base point p or the dampening λ
- ⇒ modular design, in coefficients `c` one can “plug in” any Euclidean solver for linear systems

Example I: Robust Geodesic Regression

[Fletcher, 2013; Loayza-Romero, Sibum, Welker, 2026; Baran, RB, 2026]

Given $(t_i, q_i) \in \mathbb{R} \times \mathcal{M}, i = 1 \dots, m$

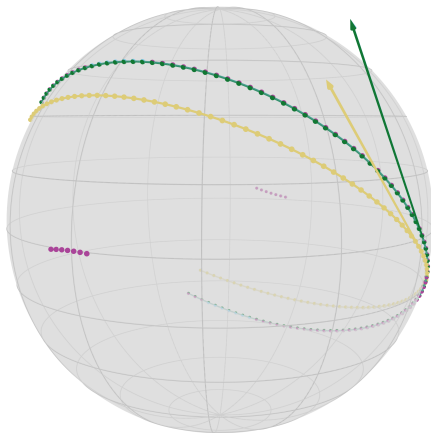
Find $\gamma_{p,X}(t) = \exp_p(tX), (p, X) \in T\mathcal{M}$
such that $\gamma_{p,X}(t_i) \approx q_i$

Objective

$$f(p, X) = \sum_{i=1}^m \rho(d^2(\gamma_{p,X}(t_i), q_i))$$

Example. $\mathcal{M} = \mathbb{S}^2$

Geodesic with **data measurement outliers**.



Remaining Error.

Least Squares	$\rho(t) = t$	1.3904×10^{-2}
Robust	$\rho_{\text{Hu}, 10^{-4}}$	2.2737×10^{-6}

Example II: The Robust Procrustes Problem

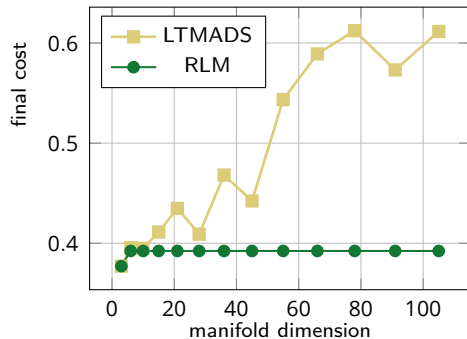
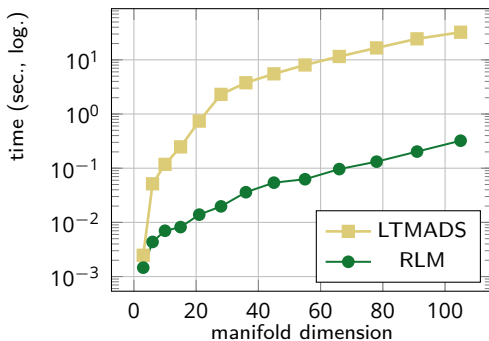
[Jasa, RB, Kümmerle, Athreya, Lubberts, 2025]

Given $A, B \in \mathbb{R}^{d \times n}$, find “best” alignment $A \approx pB$, $p \in SO(d)$.

Procrustes. $f(p) = \|A - pB\|_F$ has a closed form minimizer using an SVD.

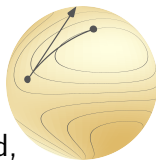
Robust. Use $\|C\|_R := \sum_{i=1}^n \|c_i\|_2$, $c_i, i = 1, \dots, n$ the columns of $C \in \mathbb{R}^{d \times n}$

Example. Data with $d = 3, \dots, 15$, $n = \frac{d(d+1)}{2}$.



Summary

List of Algorithms in Manopt.jl



Derivative-Free Nelder-Mead, Particle Swarm, CMA-ES, MADS

Subgradient-based Subgradient Method, Convex Bundle Method, Proximal Bundle Method

Gradient-based Gradient Descent, Conjugate Gradient, Stochastic, Momentum, Nesterov, Averaged, Gradient Sampling, Quasi-Newton with (L-)BFGS(B), DFP, Broyden, SR1, ...

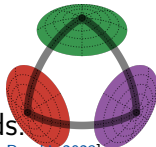
Hessian-based Trust Regions, Adaptive Regularized Cubics (ARC)

splitting Chambolle-Pock, Douglas-Rachford, Cyclic Proximal Point, Proximal Gradient, (robust) Levenberg-Marquardt

constrained Augmented Lagrangian, Exact Penalty, Frank-Wolfe, Projected Gradient, Interior Point Newton

nonconvex Difference of Convex Algorithm, DCPA

Manifolds.jl



Goal.

Efficiently implemented and well-documented Riemannian manifolds.

[Axen, Baran, RB, Rzecki, 2023]

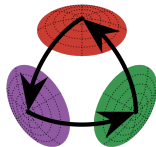
Meta. generic implementations for $\mathcal{M}^{n \times m}$, $\mathcal{M}_1 \times \mathcal{M}_2$,
vector- and tangent-bundles, esp. $T_p\mathcal{M}$

LieGroups.jl uses the manifolds provided here as a basis

Library of implemented manifolds

- ▶ Circle, Sphere, Torus, Hyperbolic, Projective Spaces, Hamiltonian
- ▶ (generalized, symplectic) Stiefel, Rotations
- ▶ (generalized, symplectic) Grassmann, fixed rank matrices
- ▶ Symmetric Positive Definite matrices, with fixed determinant
- ▶ (several) Multinomial, (skew-)symmetric, and symplectic matrices
- ▶ Tucker, Segre & Oblique manifold, Kendall's Shape space
- ▶ probability simplex, orthogonal and unitary matrices, ...

LieGroups.jl



[RB, Baran, 2026]

Goal.

Provide an interface to define and use Lie groups and provide efficiently implemented and well-documented Lie groups $\mathcal{G}$.

Meta.

Generic implementations for the Lie algebra $\mathfrak{g}$, group operations, group actions, and an identity $e \in \mathfrak{g}$ that only allocates when necessary

$$\mathcal{G}^n, \mathcal{G} \times \mathcal{H}, \mathcal{G} \ltimes \mathcal{H}, \mathcal{G} \rtimes \mathcal{H}$$

Library of implemented Lie groups

- ▶ Circle, General linear, Heisenberg, Orthogonal
- ▶ Special Euclidean, Special linear, Special Orthogonal, Special Unitary
- ▶ Translation, Unitary

Conclusion

`Manopt.jl`

- ▶ General setup of a solver and an example
- ▶ works on all Manifolds implemented with `ManifoldsBase.jl`
- ▶ ease of use, example runs & tools

Robust Riemannian Levenberg-Marquardt

- ▶ solve nonlinear least squares – robustly!
- ▶ technical differences of the Jacobian
- ▶ two examples

...and a short overview of details in `Manopt.jl`, `Manifolds.jl` & `LieGroups.jl`

Selected References



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ronnybergmann.net/talks/2026-Manoptjl-Oberwolfach.pdf